Fiscal Year:	FY 2018	**Task Last Updated:**	FY 10/20/2017
PI Name:	Gilroy, Simon Ph.D.		
Project Title:	GeneLab: Revealing Spaceflight- and Gravity-Response Networks in Plants		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Comments:	NOTE: PI formerly at Pennsylvania State University; moved to University of Wisconsin-Madison in 2007 (Info received 7/2009)		
Project Type:	Ground,NASA GeneLab	**Solicitation / Funding Source:**	2016 Space Biology (ROSBio) NNH16ZTT001N-GeneLab. Appendix A: Translational Systems Biology and Informatics Research Using the GeneLab Data System
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	25	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
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Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Barker, Richard Ph.D. (University of Wisconsin)		
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Task Description:	The GeneLab data repository is a powerful resource with which to dissect plant responses to spaceflight at the molecular and genetic level. The research in this GeneLab Innovation study therefore seeks to continue and extend our current analysis of the GeneLab plant science data through bioinformatics, experimental tests of predictions, and development of approaches to facilitate data exploration and rapid sharing of analytical tools. This work directly addresses Goal 1 of the Space Biology program by creating 'new knowledge of how biological systems adapt to space.' In addition, this research approaches some of GeneLab's guiding questions as to how implementation of large scale dataset analyses can help produce new insights and hypotheses about the effects of spaceflight on biological systems. Aim (1): We are using informatics to ask how spaceflight alters the plant transcriptome and proteome using GeneLab data. We are then comparing these changes to responses to other stresses and environments, (e.g., pathogen attack or light stress) where large-scale, publicly available datasets have already been deposited outside of GeneLab. These datasets include responses to partial gravity and gravitropism-related transcriptomes. Integrating such analyses with allied data, such as QTL maps, will allow us to specifically ask how many of the spaceflight-linked changes may be mediated through altered gravity sensing and response. Our preliminary analysis of Arabidopsis data from BRICs (Biological Research in Canisters)16-20 has revealed oxidative stress and hypoxia as major flight-related stresses and has targeted gene families ranging from heat-shock proteins to cell wall peroxidases as likely key players in this spaceflight response network. The more comprehensive analysis being undertaken in this innovation study will greatly strengthen these initial findings and expand the range of molecular mechanisms we can detect. Aim (2): The second element of the work is to take the genes defined from the informatics in Aim 1 and test their function by passing their mutants through a suite of phenotypic assays. Analyses includes morphometric measurements of gravtiropic, thigmotropic, and clinostat growth in addition to development in short and long-term hypoxia. These studies are being performed using automated image analysis tools such as RootNav, RootTrace, and custom R-Shiny apps that are highly compatible with rapid sharing of data and analyses via the cloud. Inferences from these growth analyses will be compared to those revealed by quantitative PCR results of marker genes identified from the core spaceflight, oxidative stress, hypoxia, and gravitropism transcriptomic analyses. Aim (3): GeneLab datasets also offer the opportunity to address the key challenge of facilitating exploration of these large, complex and specialized datasets by non-specialists. The Gilroy lab has a long-standing collaboration with iPlant/CyVerse and Cold Spring Harbor in the development of RNAseq analysis teaching tools such as the DNAsubway Green Line and the Cold Spring Harbor DNA learning center. Extending these approaches to the spectrum of GeneLab's plant data will make a significant step towards promoting access and understanding. One further approach to increasing accessibility for the non-specialist that we are developing will be through an online Tableau or Qlik-based app. These tools build highly interactive data visualization that allows non-expert users to search across, and explore interconnections within, large and complex datasets. The initial data visualization tool we have developed contains all GeneLab Arabidopsis data but is designed to be capable of integrating with other datasets to greatly extend the reach of the GeneLab resources.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research is leading to benefits outside of the spaceflight arena at two levels: (1) tool development and (2) extension of basic knowledge of plant stress responses. At the level of tool development, we have generated a large dataset exploration environment built around the Qlik data visualization toolset. This tool allows rapid comparison and exploration between large-scale 'omics-level datasets, facilitating comparison between transcriptomics/gene expression data and then cross-comparisons to protein-level data such as proteomics analyses. Although focused on spaceflight data within GeneLab's data repository, this architecture is applicable to wider experimental datasets. In addition, we are developing 3-d printing technologies to rapidly build plant analysis stations with a focus on large parallel arrays of clinostats and imaging stations allowing phenotyping of gravity-related effects on plants. In parallel we have also been developing gas regulation chambers to allow analysis of plant growth and molecular phenotypes related to lowered oxygen availability. Although again focused on spaceflight-related events of lowered oxygen levels, these devices are also applicable to studying the effects of ground-based phenomena such as flooding effects on plant biology. Our analyses of phenotypes has also extended basic understanding on plant responses to touch, low oxygen levels, and oxidative stress signaling on Earth. For example, we have been able to define an array of genes that when mutated alter responses to oxidative stress on Earth. Oxidative stress is a major element of stress signaling and resistance pathways and so this basic understanding is expanding our broad knowledge of how plants are able to respond to less than optimal environmental conditions.
	The major goals undertaken in this first year of the research have been to: (1) Develop a database and coupled exploration environment capable of allowing rapid exploration of the GeneLab data repository's plant science data and (2) Develop phenotyping systems to handle a relatively high throughput of plant lines to allow testing of the prediction arising from the data exploration. Progress has been made on both fronts as described below. Database: We have developed an interactive data exploration environment based on the Qlik engine that allows rapid and transparent comparison and analysis of data drawn from the GeneLab data repository. The backend database driving the analysis environment is at present focused on GeneLab's plant biology data. The raw data has been imported, reanalyzed using a standard approach across all gene expression datasets (transcriptomics data) and across all protein level analysis (proteomic data) to allow for valid comparisons. A graphical front end has then been developed to allow non-specialists to browse and analyze the datasets. The database also incorporates metadata to allow analyses to be driven in part by the kinds of experimentation and data generated and also provides links to the primary literature (where possible) to allow users to seamlessly compare their analysis to the published analyses of the same datasets. The environment has completed alpha testing and is now undergoing beta testing across several universities. A large-scale server stress test and gathering of associated user feedback is to be conducted over a few weeks in October/November 2017 to assess accessibility requirements prior to a broader public release of the initial environment. During this release of version 1, increased functionality related to incorporating other organisms into the backend database is anticipated with updates throughout the coming year as new functionality becomes available. Devices: A major limitation exists to capitalizing on the insights drawn from the kinds of data analysis outlined above. These bioinformatics-driven analyses focus in on potentially important genes for spaceflight response that will then be

	tested with experiments aimed at revealing the function of these targeted genes. These experiments revolve around making mutant lines of plants where these target genes have been deleted, and then using growth and molecular analyses to ask how these plants are now different. The mutant production pipeline is well established but the devices to allow growth and molecular analyses under conditions related to spaceflight are generally of very limited availability and can process very few lines at a time.
Task Progress:	We have therefore been working with a series of engineering students to develop more accessible devices to make these measurements on many plant lines simultaneously. We have focused on clinostats and chambers to impose altered oxygen levels as these are both poorly available technologies and directly relate to spaceflight-related stresses. The clinostat randomizes the direction of gravity, allowing analysis of effects decoupled from a coherent gravity vector. Hypoxic chambers allow the limited oxygen supply associated with spaceflight to be mimicked. Thus, hypoxia is thought to develop around respiring tissues in spaceflight as the lack of buoyancy-driven convection in microgravity limits the mixing of gasses (oxygen) around organisms. We have developed a 3-d printing approach to rapidly prototype and build devices and combined these designs with minicomputer controllers to allow precise regulation of clinostat rotation of oxygen flow for the chamber systems. These devices are in the final testing and deployment phase before a more large-scale production to support the large parallel experiment approach described above. We are already using the prototypes of these systems as a proof-of-concept that this rapid prototyping/build approach is viable to make the equipment needed to support a larger throughput of plant mutant analyses required for this research. This effort is now aimed at integrating the hypoxia chambers and clinostats to provide a unique tool to study the combined effect of these two stresses (no coherent gravity direction and low oxygen stress) as a tool for space science researchers. As part of this genetic analysis/device production strategy for capitalizing upon GeneLab data we have been able to recruit a large cohort of ~25-30 undergraduate students with approximately half having principle interests in biology and half being engineers. These students are intimately involved in device design, prototyping, and development, and in the biological analysis of the mutant plant lines. These experiences are supported through a variety of academic channels through the laboratory including volunteers, students researching as part of their in-lab independent study at the university and also as part of a class we have initiated at the University of Wisconsin called Astro-Botanical Engineering, which is focused on this blending of biological plant research with the development of novel engineering solutions to test the relevant biological hypotheses about spaceflight responses. We are extending these kinds of studies to collaborative educational efforts towards plant space biology, e.g., we have a developing collaboration with Carthage College on using this approach with their new Astrobiology class.
Bibliography Type:	Description: (Last Updated: 02/02/2026)
Significant Media Coverage	Barker R, Gilroy S. "GeneLab-related research presented at the National History Museum in the UK." Nature live at the Natural History Museum Webcast, September 2017., Sep-2017